

A linear lower bound for generalized doubly nonnegative critical exponents

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Partially resolved The general lower bound and $n = 4$ equality are verified; equality for $n \geq 5$ remains open. The dated independent agent review records the subsequent checks and supersedes original draft remarks about pending review. This is not external human peer review or formal certification.

Abstract

For every $n \geq 3$, the conventional critical exponent of the $n \times n$ generalized doubly nonnegative matrices is at least $2n - 4$. The proof uses a rational, nonnegative cyclic-bidiagonal family with two eigenvalues of order one and $n - 2$ positive eigenvalues of order ϵ . A particular entry of its real powers has a negative leading coefficient throughout the interval $(2n - 5, 2n - 4)$. Together with the published upper bound, this proves $\text{CE}_4 = 4$, resolving Conjecture 4.6 of Han, Johnson, and Paparella. Equality $\text{CE}_n = 2n - 4$ for all $n \geq 5$ is not proved. Exact rational interval certificates for eight finite examples accompany the analytic proof.

1 Statement and an explicit matrix family

A real matrix is *generalized doubly nonnegative* (GDN) if it is entrywise nonnegative, diagonalizable, and has nonnegative eigenvalues. For such a matrix and $\alpha > 0$, the conventional power is $A^\alpha = S \text{diag}(\lambda_i^\alpha) S^{-1}$. Let CE_n be the infimum of the numbers c such that A^α is entrywise nonnegative for every $n \times n$ GDN matrix and every $\alpha > c$.

Theorem 1. *For every integer $n \geq 3$,*

$$\text{CE}_n \geq 2n - 4.$$

The lower bound is already valid for GDN matrices with rational entries and distinct positive eigenvalues.

Set $m = n - 2$ and $\kappa = 1/4$. Define A_ϵ by

$$\begin{aligned} (A_\epsilon)_{ii} &= i\epsilon & (1 \leq i \leq m), \\ (A_\epsilon)_{m+1,m+1} &= 1, & (A_\epsilon)_{m+2,m+2} &= 2, \\ (A_\epsilon)_{i,i+1} &= \epsilon & (1 \leq i \leq m), \\ (A_\epsilon)_{m+1,m+2} &= 1, & (A_\epsilon)_{m+2,1} &= \kappa, \end{aligned} \tag{1}$$

with every other entry zero. For example, in order four this is

$$A_\epsilon = \begin{pmatrix} \epsilon & \epsilon & 0 & 0 \\ 0 & 2\epsilon & \epsilon & 0 \\ 0 & 0 & 1 & 1 \\ 1/4 & 0 & 0 & 2 \end{pmatrix}.$$

We shall prove that it is GDN throughout the explicit parameter range

$$0 < \epsilon \leq \frac{1}{8(m+1)}. \quad (2)$$

2 Positive, simple eigenvalues

Write

$$Q_\epsilon(z) = \prod_{i=1}^m (z - i\epsilon).$$

Expansion of the determinant of a cyclic-bidiagonal matrix gives

$$p_\epsilon(z) := \det(zI - A_\epsilon) = Q_\epsilon(z)(z-1)(z-2) - \kappa\epsilon^m. \quad (3)$$

There are just two nonzero permutation terms: the diagonal term and the full cycle. The latter has negative sign in $\det(zI - A_\epsilon)$, in every dimension.

Lemma 2. *For every ϵ satisfying (2), p_ϵ has m distinct positive roots in the respective intervals*

$$\epsilon(i - 1/4) < \lambda_i(\epsilon) < \epsilon(i + 1/4), \quad 1 \leq i \leq m,$$

and one root in each of $(3/4, 1)$ and $(2, 9/4)$. All roots are simple. As $\epsilon \downarrow 0$, the last two roots converge to 1 and 2. Moreover,

$$\lambda_i(\epsilon) = \epsilon u_i(\epsilon), \quad u_i(\epsilon) \longrightarrow u_i > 0,$$

where $u_1, \dots, u_m$ are the distinct roots of

$$q_0(u) = 2 \prod_{i=1}^m (u - i) - \kappa. \quad (4)$$

Proof. Set

$$q_\epsilon(u) = \epsilon^{-m} p_\epsilon(\epsilon u) = \prod_{i=1}^m (u - i)(1 - \epsilon u)(2 - \epsilon u) - \kappa.$$

At either endpoint $u = i \pm 1/4$, the absolute value of the product $\prod_{j=1}^m (u - j)$ is at least $3/16$. Indeed, one factor has modulus $1/4$, at most one additional factor has modulus below 1 (in which case it is $3/4$), and all remaining factors have modulus above 1. For $m = 1$ the product has modulus $1/4$.

These endpoints satisfy $0 < \epsilon u < 1/8$, so

$$(1 - \epsilon u)(2 - \epsilon u) > (7/8)(15/8) = 105/64.$$

The magnitude of the product before subtracting κ is therefore at least $315/1024 > 1/4$. Its sign changes across each interval $(i - 1/4, i + 1/4)$; subtracting $\kappa = 1/4$ does not change either endpoint sign. There is consequently a root in each of the m low intervals.

At $z = 1$ and $z = 2$, $p_\epsilon(z) = -\kappa\epsilon^m < 0$. At $z = 3/4$, each factor $3/4 - i\epsilon$ is greater than $5/8$, and thus

$$p_\epsilon(3/4) > \frac{5}{16} \left(\frac{5}{8}\right)^m - \frac{1}{4}\epsilon^m > 0.$$

The final inequality follows, for instance, from $\epsilon \leq 1/16$. Similarly, $p_\epsilon(9/4) > 0$. This provides two more positive roots. The $m + 2$ intervals are disjoint, and the polynomial has degree $m + 2$, so there is exactly one simple root in each and no other root.

For the limiting assertions, q_ϵ converges coefficientwise to q_0 . The same endpoint argument at $\epsilon = 0$ gives one simple root of q_0 in each low interval. Compactness and uniqueness then give $u_i(\epsilon) \rightarrow u_i$. Coefficientwise convergence of p_ϵ to $z^m(z-1)(z-2)$ and the two high-root intervals gives convergence of the remaining roots to 1 and 2. $\square$

Entrywise nonnegativity is apparent from (1). Lemma 2 proves that A_ϵ is GDN, with a simple positive spectrum, on the entire range (2).

3 A negative leading coefficient in a matrix power

For z off the spectrum, the $(1, m)$ entry of the resolvent is

$$((zI - A_\epsilon)^{-1})_{1m} = \frac{\epsilon^{m-1}(z-1)(z-2)}{p_\epsilon(z)}. \quad (5)$$

This is the cofactor formula along the directed path from vertex 1 to vertex m ; its $m - 1$ edges have product ϵ^{m-1} , and the two vertices outside the path contribute $z - 1$ and $z - 2$. When $m = 1$, it is the usual diagonal cofactor formula and the empty edge product is one.

Since the spectrum is positive and simple, partial fractions in (5) yield, for every real $\alpha > 0$,

$$(A_\epsilon^\alpha)_{1m} = \epsilon^{m-1} \sum_{p_\epsilon(\lambda)=0} \frac{\lambda^\alpha(\lambda-1)(\lambda-2)}{p'_\epsilon(\lambda)}. \quad (6)$$

Lemma 3. *For every fixed $\alpha > 2m - 1$,*

$$\lim_{\epsilon \downarrow 0} \epsilon^{-(2m-1)} (A_\epsilon^\alpha)_{1m} = \kappa(2^{\alpha-2m} - 1). \quad (7)$$

Proof. First consider a low root $\lambda_i = \epsilon u_i(\epsilon)$. Since

$$p'_\epsilon(\epsilon u) = \epsilon^{m-1} q'_\epsilon(u),$$

its coefficient in the spectral expansion (6), before the factor λ_i^α , is

$$\frac{(\epsilon u_i(\epsilon) - 1)(\epsilon u_i(\epsilon) - 2)}{q'_\epsilon(u_i(\epsilon))}.$$

This remains bounded: its denominator converges to $q'_0(u_i) \neq 0$. The sum of the low-root contributions is therefore $O(\epsilon^\alpha)$, which is $o(\epsilon^{2m-1})$ under the hypothesis on α .

Now let λ be one of the two high roots. The characteristic equation gives the exact identity

$$(\lambda - 1)(\lambda - 2) = \frac{\kappa\epsilon^m}{Q_\epsilon(\lambda)}.$$

Its contribution to the left side of (7), before passage to the limit, is thus

$$\frac{\kappa\lambda^\alpha}{Q_\epsilon(\lambda)p'_\epsilon(\lambda)}.$$

For the root tending to $b \in \{1, 2\}$, with c the other element of this set,

$$Q_\epsilon(\lambda) \longrightarrow b^m, \quad p'_\epsilon(\lambda) \longrightarrow b^m(b - c).$$

The two limiting contributions sum to

$$\kappa \left(\frac{1^{\alpha-2m}}{1-2} + \frac{2^{\alpha-2m}}{2-1} \right) = \kappa(2^{\alpha-2m} - 1). \quad \square$$

Proof of Theorem 1. Choose any $\alpha \in (2m - 1, 2m)$. The limit in Lemma 3 is strictly negative. Hence $(A_\epsilon^\alpha)_{1m} < 0$ for all sufficiently small positive ϵ , while Lemma 2 ensures that A_ϵ is GDN. Rational positive ϵ can be chosen arbitrarily small, so the examples may have rational entries. Since α can approach $2m = 2n - 4$ from below, the asserted lower bound follows. $\square$

Corollary 4. *The conventional critical exponent for 4×4 GDN matrices is 4.*

Proof. Theorem 1 gives $\text{CE}_4 \geq 4$. Theorem 3.3 of Han, Johnson, and Paparella [1] gives $\text{CE}_4 \leq (4^2 - 2 \cdot 4)/2 = 4$. $\square$

4 Scope, verification, and remaining question

Corollary 4 proves Conjecture 4.6 of [1]. Theorem 1 proves the lower-bound direction of the equality proposed in their Question 4.7. The upper bound $\text{CE}_n \leq 2n - 4$ for all $n \geq 5$ is not established here, so the general critical-exponent target must not be marked completely solved on the basis of this note.

The analytic argument does not depend on experiments. As an independent finite check, the accompanying standard-library verifier isolates all roots of (3) by rational sign-changing intervals and evaluates (6) with outward rational interval arithmetic. It uses the characteristic equation to replace $(\lambda - 1)(\lambda - 2)$, avoiding cancellation. At $\alpha = 2m - 1/2$, the only irrational scalar operation is $\sqrt{\lambda}$, enclosed using integer square roots. All eight enclosures below have strictly negative upper endpoint.

n	ϵ	α	$\epsilon^{-(2n-5)}(A_\epsilon^\alpha)_{1,n-2}$, approximate
3	10^{-6}	$3/2$	-0.072030414192
4	10^{-12}	$7/2$	-0.073212657290
5	10^{-18}	$11/2$	-0.073223138537
6	10^{-24}	$15/2$	-0.073223301045
7	10^{-30}	$19/2$	-0.073223304600
8	10^{-36}	$23/2$	-0.073223304700
9	10^{-42}	$27/2$	-0.073223304703
10	10^{-48}	$31/2$	-0.073223304703

The exact endpoints, not the rounded values in the table, are stored in `results/gdn_certificate.json`. From the package root, run

```
python code/gdn_certificate.py \
    results/gdn_certificate.json --verify
```

The code verifies finite examples only. It does not substitute for the all-dimensions limiting argument.

References

- [1] X. Han, C. R. Johnson, and P. Paparella, *The critical exponent for generalized doubly nonnegative matrices*, Linear and Multilinear Algebra **65**(5) (2017), 1035–1044. doi:10.1080/03081087.2016.1223009. arXiv:1407.7059v3, Theorem 3.3, Conjecture 4.6, and Question 4.7 (manuscript pp. 4 and 7–8).